INCH-POUND

MIL-C-60251A(AR)
AMENDMENT 5
26 November 1997
SUPERSEDING
AMENDMENT 4
04 November 1993

MILITARY SPECIFICATION

CARTRIDGE, 7.62MM, NATO, BALL, M80 OVERHEAD FIRE APPLICATION

MIL-C-60251A was inactivated after 6 December 1995 For New Design.

This amendment forms a part of MIL-C-60251A (MU), dated 05 March 1971, and is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete "AMCR 715-505, Volume 3" wherever it appears in the specification and substitute "SCATP-7.62".

PAGE 2

2.1: Add drawing as follows:

"9362607 - Packing and Marking for Box, Wirebound for Cartridge, 7.62mm"

PAGE 3

3.7: Delete in its entirety and substitute the following:

"3.7 Velocity. The velocity test may be performed using a velocity test barrel or the Electronic Pressure Velocity Action Time (EPVAT) System. The test method employed shall be recorded on the test results.

3.7.1 Velocity measurement by velocity test barrel. The average velocity of the sample cartridges conditioned at 70^0F $\pm$ 2^0F shall be 2,750 $\pm$ 30 ft/sec at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 20 ft/sec.

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3.7.2 Velocity measurement by EPVAT. The average velocity of the sample cartridges conditioned at 70^0F ± 2^0F shall be 2,735 ± 30 ft/sec at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 20 ft/sec."

3.8: Delete in its entirety and substitute the following:

"3.8 Chamber pressure. The chamber pressure measurement may be performed using either the copper-crush cylinder method or the EPVAT System. If the velocity is measured using the EPVAT System, the chamber pressure shall be measured using EPVAT as well.

3.8.1 Chamber pressure measurement using the copper-crush cylinder method. The average chamber pressure of the sample cartridges, conditioned at 70^0F ± 2^0F, shall not exceed 50,000 psi. Neither the chamber pressure of an individual sample cartridge, nor the average chamber pressure of the sample cartridges plus three standard deviations shall exceed 55,000 psi.

3.8.2 Chamber pressure measurement using the EPVAT System. The average chamber pressure of the sample cartridges, conditioned at 70^0F ± 2^0F, shall not exceed 365 Megapascals (52,940 psi). Neither the chamber pressure of an individual sample cartridge, nor the average chamber pressure of the sample cartridges plus three standard deviations shall exceed 400 Megapascals (58,016 psi)."

3.9: Delete "4 hours" and substitute "4 to 5 hours."

3.9.2: Delete in its entirety and substitute the following:

"3.9.2 Chamber pressure measurement at 125^0F and -65^0F using the EPVAT System. The average chamber pressure of the sample cartridges subjected to high and low temperature storage conditions shall not exceed 58,016 psi nor vary from the average chamber pressure of the sample cartridges of the same lot conditioned at 70^0F ± 2^0F by more than plus (+) 50 Mpa (7,252 psi) or minus (-) 100 Mpa (14,504 psi).

Add new paragraph 3.9.3 as follows:

"3.9.3 Velocity. The average velocity of the sample cartridges subjected to high and lot temperature storage conditions shall not vary from the average velocity of the sample cartridges of the same lot conditioned at 68^0F to 72^0F by more than plus (+) 150 or minus (-) 250 ft/sec."

3.10: Delete the word "ambient" and substitute the word "room".

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* 3.11: Delete in its entirety and substitute the following

"3.11 Port pressure. The port pressure measurement may be performed by either the copper-crush cylinder method or the EPVAT System. If the velocity is measured using the EPVAT system, the port pressure shall be measured using the EPVAT system as well.

3.11.1 Port pressure measurement using the copper-crush cylinder method. The average port pressure of the sample cartridges, conditioned for not less than 2 hours at 70^0F $\pm$ 2^0F, shall be 12,500 $\pm$ 2,000 psi.

3.11.2 Port pressure measurement using EPVAT. The average port pressure of the sample cartridges, conditioned for not less than 2 hours at 70^0F $\pm$ 2^0F, shall not be greater then 73 Mpa (10,588 psi) and shall not be less than 55 Mpa (7,977 psi)."

PAGE 8

4.3.3.1, Table II: Delete the heading "Ambient Temperature" and substitute "70^0F + 2^0F".

PAGE 9

Table II, under requirement paragraphs: Delete "3.7 & 3.9.2" and substitute "3.7 & 3.9". Delete "3.8 & 3.9.1" and substitute "3.8 & 3.9".

PAGE 11

4.3.3.2, Table III: Delete defect number "3, 4, 5, 6 and 7" and substitute "4, 5, 6, 7 and 8" respectively. Insert new defect number 3 as follows:

"3. Hangfire 7/ 0 -"

Delete defect number 3.c and substitute the following:

"3.c. Escape of gas around primer cup 6/ 56 132"

Delete "Total of defect 3c.(1), 3c.(2), and 4a(1)" and substitute "Total of defects 4.c and 5.a(1)".

PAGE 12

4.4.3.2: Add footnotes 6/ and 7/ to Table III as follows:

"6/ The total number of cases in which the escape of gas is 50% or more around the primer cup shall not exceed 18 for the first sample nor more than 41 for the cumulative sample.

7/ For definition of a hangfire, see SCATP-7.62 paragraph 10.6.2."

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4.4.8.2: Delete "for not less than 4 hours" and substitute "for 4 to 5 hours".

Add new paragraph 4.4.8.3 as follows:

"4.4.8.3 Room temperature tests. The test sample shall be conditioned for not less than 2 hours at 68^0F to 72^0F (room temperature) and fired at that temperature."

4.4.9: Delete the word "ambient" and substitute the word "room".

PAGE 14

Add new paragraph 4.4.15 as follows:

"4.4.15 EPVAT System". The EPVAT test method for chamber pressure, port pressure, velocity and action time shall be performed simultaneously at all temperatures except for action time and port pressure which are performed at 70^0F only."

5.: Delete in its entirety and substitute the following:

"5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.1.1 Packing, Level A. The cartridges shall be packed in accordance with drawing 9362607."

5.1.2 Marking. Markings shall be in accordance with Dwg. 9362607.

5.2 Shipping. When cartridges from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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